

Thesis: A Comparative Study of Large Language Models for Financial Sentiment Analysis and Their Predictive Potential for Short-Term Stock Price Movement

Component: Experiment 1 — Dataset Preparation

Description: This notebook loads and checks the Financial PhraseBank dataset (Malo et al., 2014) for use in Experiment 1.

It performs class distribution analysis and prepares the dataset for sentiment classification benchmarking across all four LLM models.

```
# Install all required libraries – Dataset Preparation
```

```
!pip install -q transformers datasets torch
!pip install -q yfinance pandas matplotlib seaborn
```

```
print("All libraries installed successfully.")
```

```
All libraries installed successfully.
```

```
# Import required libraries for dataset preparation
```

```
import pandas as pd
import numpy as np
import torch
import warnings
```

```
import yfinance as yf
import matplotlib.pyplot as plt
import seaborn as sns
```

```
warnings.filterwarnings("ignore")
```

```
# Check GPU
```

```
print(f"PyTorch version : {torch.__version__}")
print(f"CUDA available : {torch.cuda.is_available()}")
if torch.cuda.is_available():
    print(f"GPU detected : {torch.cuda.get_device_name(0)}")
else:
    print("No GPU detected")
```

```
PyTorch version : 2.10.0+cu128
CUDA available : True
GPU detected : Tesla T4
```

Add the API Key from Hugging Face using "Add New Secret" in Google Colab

```
# Connect to Hugging Face
```

```
from google.colab import userdata
from huggingface_hub import login
```

```
hf_token = userdata.get("HF_TOKEN")
login(token=hf_token)
```

```
print("Hugging Face login successful.")
```

```
Hugging Face login successful.
```

```
# Load Financial PhraseBank using the Parquet branch
```

```
# Using sentences_50agree (Full Dataset)
```

```
from huggingface_hub import hf_hub_download
```

```
file_path = hf_hub_download(
    repo_id="takala/financial_phrasebank",
    filename="sentences_50agree/train-00000-of-00001.parquet",
    repo_type="dataset",
    revision="0dd3028d70cbd18ded8887e65e83343b03a50482",
    token=hf_token
)
```

```
df_fpb = pd.read_parquet(file_path)
```

```
print("Dataset loaded successfully.")
print(f"Total sentences : {len(df_fpb)}")
print(f"Columns : {list(df_fpb.columns)}")
print(f"\nFirst 3 rows:")
print(df_fpb.head(3))
```

sentences_50agree/train-00000-of-00001.p(...): 100%

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Dataset loaded successfully.

Total sentences : 4846

Columns : ['sentence', 'label']

First 3 rows:

	sentence	label
0	According to Gran , the company has no plans t...	1
1	Technopolis plans to develop in stages an area...	1
2	The international electronic industry company ...	0

```
# Map numeric labels to text sentiment
# 0 = negative, 1 = neutral, 2 = positive
```

```
label_map = {0: "negative", 1: "neutral", 2: "positive"}
df_fpb["sentiment"] = df_fpb["label"].map(label_map)
```

```
# Check class distribution
print("=" * 45)
print("Financial PhraseBank - Class Distribution")
print("=" * 45)
```

```
counts = df_fpb["sentiment"].value_counts()
total = len(df_fpb)
```

```
for sentiment in ["positive", "negative", "neutral"]:
    count = counts[sentiment]
    print(f" {sentiment:<12}: {count:>5} ({count/total*100:.1f}%)"
```

```
print(f" {'TOTAL':<12}: {total:>5}")
print("=" * 45)
```

```
=====
Financial PhraseBank - Class Distribution
=====
```

```
positive   : 1363 (28.1%)
negative   : 604 (12.5%)
neutral    : 2879 (59.4%)
TOTAL      : 4846
=====
```

```
# Print 5 sample sentences per class to inspect quality of labels
```

```
for sentiment in ["positive", "negative", "neutral"]:
    print(f"\n— 5 samples: {sentiment.upper()} —")
    samples = df_fpb[df_fpb["sentiment"] == sentiment]["sentence"].sample(5, random_state=42)
    for i, s in enumerate(samples, 1):
        print(f" {i}. {s}")
```

```
— 5 samples: POSITIVE —
```

1. The new agreement , which expands a long-established cooperation between the companies , involves the transf
2. (ADP News) – Finnish handling systems provider Cargotec Oyj (HEL : CGCBV) announced on Friday it won ord
3. The world 's biggest magazine paper maker said the program to improve efficiency will include closing severa
4. a January 11 , 2010 EPHC board of directors has approved an increase in the quarterly dividend from \$ 0.03 t
5. With this appointment Kaupthing Bank aims to further co-ordinate Capital Markets activities within the Group

```
— 5 samples: NEGATIVE —
```

1. The company decided at the end of 2008 to temporarily shut down its ammonia plant in Billingham and extend t
2. down to EUR5 .9 m H1 '09 3 August 2009 – Finnish media group Ilkka-Yhtymä Oyj (HEL : ILK2S) said today its
3. The steelmaker said that the drop in profit was explained by the continuing economic uncertainty , mixed wit
4. Finland-based Stockmann Group has closed seven franchising sports stores Nike in Russia .
5. Operating loss before non-recurring items was EUR 0.9 mn , compared to a profit of EUR 11.5 mn in 2008 .

```
— 5 samples: NEUTRAL —
```

1. The center offers a comprehensive range of device design services spanning from electronics , mechanics and
2. AffectoGenimap builds highly customised IT solutions for its customers in Finland and the Baltic countries .
3. The bank 's leasing arm Nordea Liising ended the year with a profit of 4.4 million euros .
4. Her present position is the director of Stockmann 's international department stores .
5. Ahlstrom 's 5,800 employees serve customers via sales offices and production facilities in more than 20 coun

```
# Sentiment label distribution chart – Financial PhraseBank
```

```
sentiments = ["Positive", "Negative", "Neutral"]
counts_ordered = [1363, 604, 2879]
colors = ["green", "red", "blue"]
```

```
fig, ax = plt.subplots(figsize=(8, 5))
```

```
bars = ax.bar(sentiments, counts_ordered, color=colors,
              width=0.5, edgecolor="white", linewidth=0.8)
```

```
for bar, count in zip(bars, counts_ordered):
    ax.text(
        bar.get_x() + bar.get_width() / 2,
        bar.get_height() + 30,
        str(count),
        ha="center", va="bottom",
        fontsize=11, fontweight="bold"
    )

ax.set_xlabel("Sentiment Class", fontsize=12, labelpad=10)
ax.set_ylabel("Number of Sentences", fontsize=12, labelpad=10)
ax.set_ylim(0, 3400)
ax.spines["top"].set_visible(False)
ax.spines["right"].set_visible(False)
ax.tick_params(axis="both", labelsize=11)

plt.tight_layout()
plt.show()
```

